

# Camera-robot calibration test protocol

Dishwasher robot

December 9, 2025

## 1 Person doing the test

Name:

Mail:

## 2 Experiment:

Hardware      Software

### 2.1 Reason for experiment

Enter reason:

### 2.2 step by step

- Run the calibration program
- Note the positions used
- Save the data/statistics from the program
- Fill out the document with the data and your own notes

### 3 Result

Commit ID:

Which calibration points were used. Position 1-10 belongs to zone 1, 11 - 20 to zone 2 etc. Fill in a checkbox even if not all positions in the zone was used.

Zone 1:      Zone 2:      Zone 3:      Zone 4:

| Metric                          | Value | Description                                                                                                                                     |
|---------------------------------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| RMSE                            |       | Root Mean Square Error, the difference between the robots coordinates and the cameras coordinates after transformation to the robots frame.(mm) |
| STD                             |       | Standard deviation error between robot coordinates and transformed camera coordinates.(mm)                                                      |
| Number of calibration positions |       | How many calibration positions were used.                                                                                                       |
| Camera distance to robot base   |       | Measured distance between camera, x-y plane, and robot base, absolute value.(mm)                                                                |
| Camera height                   |       | Measured height between camera and the robot base, absolute value.(mm)                                                                          |
| Estimated camera position       |       | The rough angle of the cameras position to the robot base, e.g. 150 degrees, left of robot. (degrees)                                           |
| Estimated camera pose           |       | The estimated angle of camera to the robot base, e.g. z = -20, y = -10 (degrees)                                                                |

Table 1: Result table

Notes:

Clear all